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May 4, 1965
SUPERSEDING
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August 3, 1961

FEDERAL SPECIFICATION

FILM, THERMAL DEVELOPING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers the requirements for thermal developing film supplied in rolls, in a range of widths from 16mm to 9½ inches. The duplicating film specified herein is of a type in which the image is formed by light sensitive elements in a plastic vehicle coated on a flexible transparent, dimensionally stable base.

1.2 **Classification.**

1.2.1 *Types, subtypes, classes, and kinds.* The thermal developing film covered by this specification shall be of the following type, subtypes, classes, and kinds.

Type

I—Duplicating

Subtypes

A—Nonreversible

B—Reversible

Classes

1—Low Gamma

2—Medium Gamma

Kinds

1A—2 mil

1B—3 mil

1C—4 mil

1D—5 mil

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 **Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L–F–330—Film, Photographic, Black and White, General Specification for.

PPP–B–636—Box, Fiberboard.

PPP–T–60—Tape, Pressure Sensitive Adhesive, Waterproof for Packaging and Sealing.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

FSC 6750

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Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105—Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129—Marking for Shipment and Storage.

MS–26565—Spool—Photographic Film, Aerial.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

American Standards Association Inc., (ASA), Standards:

PH1.25–1956—Safety Photographic Film, Specification for.

PH2.19–1959—Diffuse Transmission Density.

PH5.3–1958—16mm. and 35mm. Microfilms on Reels or in Strips, Specifications for.

PH1.35–1961—35mm. 100-Foot Spools for Recording Instruments, Microfilm and Still Picture Cameras, Dimensions for.

PH1.13–1953—Dimensions for Molded Type Cores for Photographic Film and Paper Rolls.

PH1.20–1963—70mm. Unperforated and Perforated Film for Cameras Other Than Motion Picture Cameras, Dimensions for.

(Copies may be obtained from the American Standards Association, 10 East 40th Street, New York, N.Y. 10016.)

American Society for Testing and Materials (ASTM) Standards:

D–882–56T—Methods of Tests for Properties of Thin Plastic Sheets and Films.

D–1004–49T—Methods of Test for Initial Tear Strength.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

Technical Association of Pulp and Paper Industry (TAPPI) Standard:

T411–M—Thickness and Density of Paper.

(Copies may be obtained from the Technical Association of the Pulp and Paper Industry, 360 Lexington Avenue, New York 17, New York.)

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3. REQUIREMENTS

3.1 Materials. The thermal developing film procured in accordance with this specification shall meet all the requirements herein specified. The base film shall be composed of virgin materials. The working surface shall contain no broken areas, surface defects, or other defects of manufacture.

3.1.2 *Film base.* The film base shall be a polyester film which is an optically homogenous, clear, transparent, dimensionally stable medium and shall be noncolor selective and supplied perforated or nonperforated as specified (see 6.1.1).

3.1.2.1 *Perforations.* When specified, perforated film shall have type II perforations in accordance with ASA Standard PH1.20–1963.

3.1.3 *Safety base.* The base material shall meet the safety limits as defined in ASA Standard PH1.25–1956, "Safety Photographic Film".

3.1.3.1 *Spools and cores.* Unless otherwise specified all types, subtypes, classes and kinds of film, specified herein, shall be supplied on standard spools or cores. For 16mm. and 35mm. film widths the spools shall conform to ASA Standard PH1.35–1961. For 70mm. through 9½ inch film widths the spools shall conform to MS–26565. Cores, when specified, shall conform to ASA Standard PH1.13–1953.

3.1.4 *Tensile strength.* The tensile strength of the base film shall be not less than that shown in table I when tested as specified in 4.3.2.

TABLE I.

Tensile strength (minimum)	Tear (minimum) per inch of thickness
10,000 p.s.i.	1,000 pounds

3.1.4.1 *Initial tear strength.* Initial tear strength of the base film shall be not less than that shown in table I when tested as specified in 4.3.3.

3.1.5 *Widths.* Roll film shall be slit in accordance with table II. Dimensions shown apply to the material immediately after slitting.

TABLE II—*Slitting widths and tolerances*

Nominal film width	Film width tolerance (in inches)
16mm	0.628 ± 0.001
35mm	1.374 ± 0.002
70mm	2.754 ± 0.002
3¼ inches	3.25 ± 0.005
5 inches	4.960 ± 0.010
7 inches	6.991 ± 0.010
9½ inches	9.460 ± 0.010

3.1.6 *Lengths.* The film lengths shall be as designated in table III. The total length furnished shall include both the film length and the "total integral self-leader and trailer" quantities.

3.1.7 *Thickness.* The thickness of the film before processing, with coating included, shall be as follows:

Kind	Base thickness plus coating (in inches)
1A	0.0024±0.0005
1B	0.0034±0.0007
1C	0.0044±0.0008
1D	0.0054±0.0010

3.1.8 *Coating defects.* The film shall be free from scratches, pinholes, blisters, tears, fingerprints or other physical defects; and from streaks, spots or areas of marked higher or lower sensitivity than surrounding portions when tested as specified in 4.3.5.

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TABLE III—*Film lengths*

Film width	Film length (feet)*	Total integral self-leader and trailer (feet)*
16mm	100	6
16mm	200	6
16mm	300	10
16mm	400	15
16mm	500	15
16mm	1000	20
35mm	100	6
35mm	200	6
35mm	300	10
35mm	400	15
35mm	500	15
35mm	1000	20
70mm	100	10
70mm	200	15
70mm	300	20
70mm	400	25
70mm	500	25
70mm	1000	30
3¼ inch	100	10
3¼ inch	200	15
3¼ inch	300	20
3¼ inch	400	25
3¼ inch	500	25
3¼ inch	1000	30
5 inch	100	10
5 inch	200	15
5 inch	300	20
5 inch	400	20
5 inch	500	20
5 inch	1000	30
7 inch	100	10
7 inch	200	15
7 inch	300	20
7 inch	400	20
7 inch	500	20
7 inch	1000	30
9½ inch	100	10
9½ inch	200	15
9½ inch	300	20
9½ inch	400	20
9½ inch	500	20
9½ inch	1000	30

* Dimensions shown are minimum requirements.

3.1.9 *Thermal coefficient of expansion.* The thermal coefficient of expansion shall be as shown in table IV. The coefficient of thermal expansion shall not exceed 0.000015 inch per inch for each °F. change in temperature in the range of 70° to 140°F. when tested as specified in 4.3.6.

TABLE IV—*Dimensional stability*

Coefficient of thermal expansion	Coefficient of hygroscopic expansion	Permanent change	
		Temperature	Humidity
0.000015 inch (per inch degree F.)	0.000012 inch (per inch percent RH)	Cycle 0.10%	Cycle 0.10%

3.1.10 *Hygroscopic coefficient of expansion.* The hygroscopic coefficient of expansion shall be as shown in table IV. The coefficient of hygroscopic expansion shall not exceed 0.000012 inch per inch for each percent change in the relative humidity in the range of 20 percent to 90 percent at 70°F. when tested as specified in 4.3.7.

3.1.11 *Permanent change in linear dimensions.* The permanent change in linear dimensions shall not exceed 0.10 percent when tested as specified in 4.3.8 and 4.3.9.

3.2 Sensitometric characteristics.

3.2.1 *Subtype A, classes 1 and 2.* Subtype A, classes 1 and 2 film shall meet the sensitometric characteristics shown in table V. Subtype A duplicating film shall produce negative images from positive images, or positive images from negative images.

3.2.2 *Subtype B, class 2.* Subtype B, class 2 film shall meet the sensitometric characteristics shown in table V. Subtype B duplicating film shall produce a negative image from a negative image or a positive

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image from a positive image through reversal procedure and process.

3.2.3 *Density.* Except for the actinic background density, all other densities specified herein shall be visual diffuse transmission densities in accordance with PH2.19–1959. The maximum density shall be not less than that shown in table V when tested as specified in 4.3.10. The minimum density shall not exceed that shown in table V when film is exposed and processed in accordance with the manufacturers instructions.

3.2.4 *Actinic background density.* Actinic background density shall not exceed 0.15 density units when tested in accordance with 4.3.11.

3.2.5 *Gamma.* Gamma shall be as specified in table V. Gamma shall be as measured and defined in L–F–330.

3.2.6 *Resolution.* Resolution of all film specified herein shall be not less than the lines per millimeter shown in table V, when tested as specified in 4.3.13.

3.2.7 *Speed rating.* Speed rating shall meet the values shown in table V when tested as specified in 4.3.10.1, 4.3.14, 4.3.15, and 4.3.16.

3.3 Expiration date. Unless otherwise specified herein, the expiration date shall be not less than five years from date of shipment when the film is stored in the original unopened packages in a room maintained at a temperature not higher than 73°F. and 50 percent relative humidity. The expiration date shall be shown on the package.

3.3.1 *Radiation damage.* When subjected to ionizing radiation below 100,000 roentgens neither the chemical nor physical properties of the thermal developing film shall be so affected as to alter the sensitometric characteristics of the emulsion or to hinder the use of the film in equipment with which it is normally employed.

3.4 Uniformity of characteristics. Uniformity of physical and sensitometric characteristics among specimens of the same manufacture and coating number, and among different manufacturing lots of a given type, class, and speed shall be maintained within closest practical limits.

3.5 Workmanship. Workmanship shall be in accordance with high grade commercial practice covering the manufacture of heat developing film sensitized by a coating of light sensitive elements in a plastic vehicle.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as other-

TABLE V—*Sensitometric characteristics*

Type	Subtype	Class	D Max. (minimum)	D Min. (maximum)	Gamma	Speed-rating	Resolution lines /MM
I	A	1	0.45	0.15	0.30±0.08	1.10±0.15	144
I	A	2	0.45	0.15	0.47±0.10	0.80±0.15	161
I	B	2	0.40	0.22	0.40±0.15	0.70±0.20	144

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wise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with provisions set forth in MIL–STD–105, except where otherwise indicated hereinafter. For purposes of sampling, an inspection lot for examination and tests shall consist of all materials of the same type and class submitted for delivery at one time.

4.2.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.2 *Inspection of the end item.*

4.2.2.1 *Examination of the end item.* Examination of the end item shall be made in accordance with the classification of defects, set forth below and inspection levels and acceptable quality levels (AQLs) specified in 4.2.2.1.5. Not more than 3 rolls of roll film shall be drawn from any one shipping container for each lot of film offered for acceptance for visual and dimensional characteristics. The lot size for purposes of determining sample size in accordance with MIL–STD–105 shall be expressed in units of rolls for examination under 4.2.2.1.1, 4.2.2.1.2, and 4.2.2.1.3 and in units of shipping containers for examination in 4.2.2.1.4. Examinations included in 4.2.2.1.1, 4.2.2.1.2, and 4.2.2.1.3 may be performed at any point immediately before, during or after the spooling or packaging operation.

4.2.2.1.1 *Examination of the end item for defects in appearance and workmanship.* The sample unit for this examination shall be one linear foot of sensitized film as applicable. Not more than three sample units shall be drawn from any one roll.

Examine	Defect
Appearance and workmanship	Film not completely or evenly coated.
	Emulsion surface shows scuffs, scratches, foreign matter, or abrasions.
	Holes, pinholes, tears, cuts, cracks, fingermarks, streaks, spots, or sharp creases.

4.2.2.1.2 *Examination of the end item for dimensional defects.* The sample unit for this examination shall be one roll of film or one linear foot of film as applicable.

Examine	Defect
Width	Not within tolerances.
Thickness	Not within tolerances.

4.2.2.1.3 *Examination of the end item for defects in construction and length and blocking.* The sample unit for this examination shall be one roll of roll film.

Examine	Defect
Roll	Spool (when required) not specified type, damaged. Film unevenly or improperly wound; less than specified number of feet. Roll not continuous. Roll contains splice.

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4.2.2.1.4 *Examination of preparation for delivery.* An examination will be made to determine that marking, materials, workmanship, contents, and weights comply with the requirement of section 5 of this specification. Defects will be scored in accordance with the classification below. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Examine	Defect
Packaging and preservation	Not as specified, not in accordance with contract requirements. Rolls not unit wrapped and packaged as specified. Unit packages not uniform. Containers of improper construction.
Packing	Not packed as specified. Packing material not as specified; closures not accomplished by specified or required methods or materials.
	Number of rolls per carton less than indicated or specified.
Markings (interior and exterior)	Marking not in accordance with contract specifications. Labels when used not properly attached. Markings illegible, incorrect, incomplete, omitted, or not in accordance with contract requirements.
Weight (interior and exterior)	Gross weight exceeds contract requirements.

4.2.2.1.5 *Inspection levels and acceptable quality levels (AQLs) for examinations.* The inspection levels, for determining the sample size, and the acceptable quality levels (AQLs), expressed in defects per 100 units, shall be as follows:

Examination	Inspection level	AQL
4.2.2.1.1	I	1.0
4.2.2.1.2	S–3	2.5
4.2.2.1.3	S–2	2.5
4.2.2.1.4	S–4	4.0

4.2.3 *Testing of the end item.* Testing of the end item shall be conducted in accordance with table VI for the characteristics as indicated therein on each lot submitted for acceptance. The sample unit, as referenced herein is defined as the amount of material required to perform all tests one time. The sample unit for all testing shall consist of one roll of film. Three sample units, each drawn from a different shipping container, shall be tested with no evidence of failure to meet sample unit or average requirements, as applicable. As an alternate, samples for testing for base density may be drawn at any points after the base casting or extrusion operation. Samples for testing for the sensitometric characteristics, may be drawn at any points after the emulsion coating operation.

4.3 Test methods.

4.3.1 *Physical tests.* The following tests shall be made in accordance with the applicable test methods specified in American Society for Testing and Materials, and the Technical Association of the Pulp and Paper Industry.

Test	*Method No.*
Tensile strength	D–882–56T
Initial tear strength	D–1004–49T
Thickness	TAPPI–T411–M

4.3.2 *Tensile strength.* Tensile strength shall be tested in accordance with ASTM method D–882–56T, or the latest revision, when the material is at 75°F.

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TABLE VI—*Instructions for testing (sample unit)*

Characteristic	Specification reference		Rqmts. appl to— Indiv. unit	Number determinations per sample unit	Results reported as—	
	Requirement	Test method			Pass or fail	Numerically to nearest
Resolution	3.2.4	4.3.13	X	Avg. 5	X	—
Sensitometric property	3.2.1 (Table V)	4.3.10	X	Avg. 2	—	0.01 scale
Permanent change in linear dimension	3.1.10	4.3.8 4.3.9	X	Avg. 5	—	.01 percent

4.3.3 *Initial tear strength.* Initial tear strength shall be tested in accordance with ASTM method D–1004–49T or latest revision.

4.3.4 *Thickness.* Thickness of the material, with or without coating shall be measured using the methods specified in TAPPI-T411–M.

4.3.5 *Coating defects.* Two samples of each kind of film as specified herein, not less than 12 inches each in length, shall be selected at random from the film to be tested. One half of the samples shall be put aside and protected from exposure to light and processing. The unprotected half shall be examined in visible light to determine conformance with requirements specified in 3.1.8. The protected half of the samples shall be exposed and fully processed on applicable equipment. These samples (after exposure and processing) shall be held not more than six feet from an unshielded 60-watt incandescent lamp and again visually examined for compliance with requirements specified in 3.1.8.

4.3.6 *Thermal coefficient of expansion.* Cut a sample, not less than 24 inches in length very accurately. Normalize the sample at 70° ± 2° F. and 50 percent ± 2 percent relative humidity for twenty-four hours and immediately measure the sheet with accurate graduated calipers or steel rule. Locate the exact positions where the measurements were taken. Place the sample in a controlled container having a temperature of 120° ± 2° F. and a constant relative humidity of 50 percent ± 2 percent. Maintain this condition for twenty-four hours and measure the test sample at the locations previously marked and record the change. The thermal coefficient of expansion is calculated as follows:

$$\text{Thermal coefficient of expansion} = \frac{D_p - D_o}{50 \times D_o}$$

D_o = original length (or width) before heat aging.

D_p = final length (or width) after aging.

50 = difference in temperature (120° — 70°F.).

4.3.7 *Hygroscopic coefficient of expansion.* Cut a sample, not less than 24 inches in length very accurately. Normalize the sample at 20 percent ± 2 percent relative humidity and 72° ± 2°F. for twenty-four hours and immediately measure the sheet

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with accurate graduated calipers or steel rule. Locate the exact position where the measurements were taken. Place a sample in a controlled container having a relative humidity of 90 percent ± 2 percent and a constant temperature of 72° ± 2°F. Maintain this condition for twenty-four hours and measure the test sample at the locations previously marked and record the change. The hygroscopic coefficient of expansion is calculated as follows:

$$\text{Hygroscopic coefficient of expansion} = \frac{D_f - D_o}{70 \times D_o}$$

D_o = original length (or width) before humidity aging.

D_f = final length (or width) after humidity aging.

70 = difference in relative humidity (90 percent — 2 percent).

4.3.8 *Permanent change, temperature cycle.* Cut five samples, 24 inches each in length very accurately, and mark the midpoints of each edge for use as reference points. Condition the five samples at 70° ± 2° F. and 50 percent relative humidity ± 2 percent relative humidity for a period of twenty-four hours. At the end of the conditioning period (24 hours) measure and record the distance between the opposite edge of the test samples at the reference marks to the nearest 0.01 inch. Place the samples for one hour in a circulating air oven heated to 180° ± 2° F. At the end of the oven exposure period allow the specimens to cool for 24 hours to the original conditioning temperature and relative humidity. Measure and record the distance between the opposite edges of the test samples at the reference marks to the nearest 0.01 inch. The percent change is calculated as follows:

$$\text{Percent change} = \frac{D_o - D_f}{D_o} \times 100$$

D_o = original length (or width) of test sample in inches.

D_f = final length (or width) of test sample in inches.

Average the values obtained for each length and width direction and report as the permanent change, temperature cycle.

4.3.9 *Permanent change, humidity cycle.* Cut five samples, 24 inches each in length very accurately and mark the midpoints of each edge for use as reference points. Condition the five samples at 70° ± 2° F. and 20 percent ± 2 percent relative humidity for a period of twenty-four hours. At the end of the conditiong period (24 hours) measure and record the distance between the opposite edges of the test samples at the reference marks to the nearest 0.01 inch. Expose the samples to a relative humidity of 90 percent ± 2 percent at 70° ± 2° F. for a period of twenty-four hours, then condition the sample for 24 hours at 20 percent ± 2 percent relative humidity and 70° ± 2° F. Measure and record the distance between the opposite edges of the test samples at the reference marks to the nearest 0.01 inch. The percent change is calculated as follows:

$$\text{Percent change} = \frac{D_o - D_f}{D_o} \times 100$$

D_o = Original length (or width) of test samples in inches.

D_f = Final length (or width) of test samples in inches.

Average the values obtained for each length and width direction and report as the permanent change, humidity cycle.

4.3.10 *Maximum density.*

4.3.10.1 *Subtype A.* The maximum density shall be measured on a sample piece of

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film which is first exposed on a calibrated Kalvar Standard Sensitometer, Model A, or a sensitometer of equal or better accuracy. The sample shall be exposed for 20 seconds and then processed in accordance with the manufacturer's instructions.

4.3.10.2 *Subtype B.* The film sample, in contact with the carbon step wedge #5 specified in 4.3.14, shall be exposed for 4 seconds with the calibrated AH–6 light source specified in 4.3.13. The exposed film shall then be treated in a Gass Diffusion Oven for 2½ minutes at a temperature of 150° ± 2°F. At the end of this treatment period the film shall be re-exposed, without the step tablet, for 6 seconds with the AH–6 light source. After this re-exposure the film shall be final processed on the Kalvar Developer, Model 6, at a temperature of 255° ± 5°F.

4.3.11 *Actinic background density.* The actinic background density shall be no greater than 0.15 density units. The actinic background density shall be measured, using a Welch Densichrone, Model No. 3835, using the blue filter supplied with the instrument, or an instrument of equal or better accuracy.

4.3.12 *Gamma, classes 1 and 2.* The gamma shall be determined by measuring the slope of the straight line portion of an H and D curve made from a step wedge exposed for 20 seconds on the Kalvar Standard Sensitometer, Model A, or a sensitometer of equal or better accuracy and developed in accordance with the manufacturer's instructions.

4.3.13 *Resolution.* A General Electric AH–6 water cooled mercury vapor lamp or a lamp with equal or better characteristics shall be used for exposure. The lamp shall be a distance of approximately ten inches from the film plane and the exposure is to be made using a vacuum frame. Exposure times in excess of twenty seconds should not be used. The maximum density obtained in the sample being tested should be between 0.40 and 0.55. A silver resolving power test target as manufactured by Backbee-Mears Company, St. Paul, Minn., or equivalent is to be used. The diazo target made from the silver resolving power test target should have a density in the unexposed areas of at least 1.5 when measured on a Welch Densichrone No. 3835, using the blue filter. The film being tested should be examined for resolution using an 80 to 100 microscope, and the last group resolved should be the last group in which it is possible to count the individual lines in each direction.

4.3.14 *Apparatus for determining speed rating.* The following apparatus, or other appropriately designed equipment, shall be used in determining speed rating (4.3.16). The apparatus is obtainable at the corresponding listed sources.

(1) Kalvar Sensitometer
Kalvar Corporation
909 South Broad Street
New Orleans 21, Louisiana

(2) Photoelement, Dr. B. Lang 550 No. 103
Epic Incorporated
154 Nassau Street
New York 38, New York

(3) Schott Filter No. UG1, 2mm, thick
Fish-Schurman Corporation
70 Portman Road
New Rochelle, New York

(4) Weston Milliammeter, D.C., Model No. 301
Weston Electric Instruments Corporation
Newark 5, New Jersey

(5) Eastman Carbon Step Wedge, No. 5, Calibrated
Eastman Kodak
Rochester, New York

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(6) Kalvar Developer, Model No. 6
Kalvar Corporation
909 South Broad Street
New Orleans 21, Louisiana

(7) Welch Densichrone, Catalogue No. 3830A
Light Box, Catalogue No. 3853A
W. M. Welch Manufacturing Co.
1515 Sedgwick Street
Chicago 10, Illinois

(8) Kalvar Gass Diffusion Oven
Kalvar Corporation
909 South Broad Street
New Orleans 21, Louisiana

4.3.15 *Sensitometer calibration.* Using the Lange cell, filter and milliammeter, center the cell in the center of the exposure window of the sensitometer. With the lamp on and open shutter, record the milliammeter reading. Compare the reading to the calibration curve of the cell provided by Kalvar Corporation. If the reading is different, vary the voltage input to the sensitometer lamp until the cell reading is correct. Wait five minutes and repeat until there is no change in the cell reading on repeated tests. Record the reading on the sensitometer ammeter. Each time the sensitometer is turned on, wait ten minutes, open shutter and adjust the lamp voltage to give this correct meter reading.

4.3.16 *Speed rating test procedure.* Turn on the sensitometer, Kalvar Developer and Densichrone. Wait fifteen minutes. Check the open shutter reading on the sensitometer ammeter and adjust to the correct value with powerstat. Close the shutter. Check the temperature of the developer and adjust to 230° to 240° F. Set the sensitometer timer on 20. Tape the Kodak density strip on the sensitometer window, dull side up. Place the test film strip, emulsion down, in contact with the Kodak density strip and fasten down the contact belt. Open the shutter. When the shutter closes, remove the test film strip immediately and develop in the Kalvar Developer. Zero the densichrone and measure the film strip in the transparent area. Record this value. Add 0.10 to this value and find the step in the film strip which is nearest to this sum. Record this density and step number. Find the step in the Kodak density strip corresponding to this step in the test film strip. Record the value of this Kodak density strip step from its calibration table. Repeat this procedure with two more samples. Average the results and record this average value as the "speed figure" of the film.

5. PREPARATION FOR DELIVERY

(For civil agency procurement, the levels of preservation, packaging, and packing used are defined in Fed. Std. No. 102.)

5.1 Packaging.

5.1.1 *Level A.* The film shall be packaged in a metal, light-tight water vapor-resistant container of suitable size and design. The metal container shall be adequately sealed against moisture and light with a water-resistant tape conforming to PPP–T–60.

5.1.2 *Level C.* Films shall be packaged in accordance with the contractor's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Film preserved and packaged to meet 5.1.1 shall be packed in exterior-type shipping containers in accordance with PPP–B–636, type CF, class weather-resistant, V3c. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required and contain identical quantities. The gross weight of each pack shall be limited to approximately 90 pounds. Container strength and design shall be adequate to insure protection against damage during the shipment, handling, intermediate storage and worldwide redistribution. Containers shall be

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closed in accordance with the specification appendix. Strapping shall not be required.

5.2.2 *Level B.* Film packaged to meet 5.1.1, shall be packed in exterior-type shipping containers in accordance with PPP–B–636, type CF, class domestic. As far as practical, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. The gross weight of each pack shall be limited to approximately 90 pounds. Container strength and design shall be adequate to insure protection against damage during multiple domestic shipments, handling, and covered storage.

5.2.3 *Level C.* Items preserved and packaged as specified, shall be packed to insure carrier acceptance and safe delivery to destination at the lowest transportation rate. Shipping containers, as a minimum, shall comply with carrier rules and regulations applicable to mode of transportation.

5.3 **Marking.**

5.3.1 *Military agencies.* All shipping containers shall be marked in accordance with MIL–STD–129, except that the dimensions of the exterior container need not be included. The nomenclature shall be:

FILM PHOTOGRAPHIC, THERMAL DEVELOPING

The type, subtype, size (width and length), expiration date, emulsion or coating numbers shall be clearly marked on the containers.

5.3.2 *Civil agencies.* The following information shall be included in the labeling provided: Manufacturer's name, trade name of product, width and length, coating or emulsion number, type, subtype and kind, expiration date, exposure index, the name "Film, Photographic, Thermal Developing," perforated or nonperforated, specification number and stock number. Labels shall be attached in such a manner that they will not be damaged when the package is opened. Unless otherwise specified, exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.3 The following precautionary marking shall be permanently and legibly marked in bold contrasting color such as red: "FILM, PHOTOGRAPHIC, THERMAL DEVELOPING", "STORE IN COOL LOCATION".

6. NOTES

6.1 Intended use. The films covered by this specification are intended for use in printing and development equipment of a type recommended by the manufacturer for the reproduction of negative or positive image prints from negative or positive image film masters as shown below.

From	To
a. Negative image	Positive image
b. Positive image	Negative image
c. Negative image	Negative image
d. Positive image	Positive image

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Stock number.

(*c*) Type, subtype, class and kind.

(*d*) Width and length.

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(*e*) Selection of samples.

(*f*) Levels of packing and packaging.

(*g*) Preproduction samples are or are not required.

(*h*) Perforated or nonperforated.

6.3 Guarantee. Upon request, the manufacturer shall replace without cost to the Government, any material which exhibits latent or other physical defects prior to the expiration date.

6.4 Cross reference of classification. Paragraph 1.2.1 of the superseded basic specification has been completely revised in this specification. The items covered by this specification may be binned with prior items. The difference in classification between the basic superseded specification and this specification are shown in the cross reference below:

Classification	Basic specification	This specification
Type	Heat developing	Type I—Duplicating
Subtypes	None	A—Nonreversible B—Reversible
Classes	A—2 mil base B—3 mil base C—5 mil base	1—Low Gamma 2—Medium Gama
Kinds	None	1A—2 mil 1B—3 mil 1C—4 mil 1D—5 mil

6.5 Definitions.

6.5.1 *Kalvar Standard Sensitometer, Model A.* A sensitometer designed and developed for precise and reproducible measurements of the ultra-violet-light sensitivity of heat developing materials which are coated with light sensitive elements in a plastic vehicle. These light sensitive elements are capable of forming light scattering centers upon exposure to actinic radiation and upon the subsequent application of heat.

6.5.2 *Light scattering.* This term is used to describe the method of image formation in films in which opacity is obtained by the creation of light scattering elements which reflect and diffract light incident upon the film as opposed to those films in which the incident light is attenuated by absorption. Such light scattering films are characterized by a very marked difference between the visual diffuse transmission density, the printing diffuse transmission density and the projection density. The printing diffuse transmission density range and the projection density range are much greater than the visual transmission range.

6.5.3 *Visual diffuse transmission density.* As defined in ASA Standard PH2.19–1959.

6.5.4 *Printing diffuse transmission density.* As defined in ASA Standard PH2.19–1959.

6.5.5 *Maximum density.* The visual diffuse transmission density obtained when a light scattering film is exposed sufficiently to decompose all of the sensitizer and when this piece of film is processed according to the manufacturer's instructions.

6.5.6 *Speed.* Speed is a comparative measure of exposure to light required to produce a correctly exposed image.

6.6 Transportation description. Transportation description and minimum weights applicable to this commodity are:

Rail:

Films, not otherwise indexed by name, unexposed.

Carload minimum weight 30,000 pounds.

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Motor:

Films, unexposed, not otherwise indexed.

Truckload minimum weight (w) 30.6 pounds, (w) subject to Rule 115, National Motor Freight Classification.

MILITARY CUSTODIANS:

Navy—WP

Army—EL

Air Force—88

Preparing Activity:

Navy—WP

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Copies of this specification may be purchased for 10 cents each.

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